```
geetha@Geetha:~$ mkdir os
geetha@Geetha:~$ cd os
geetha@Geetha:~/os$ nano lab6.sh
geetha@Geetha:~/os$ chmod +x lab6.sh
geetha@Geetha:~/os$ ./lab6.sh
Lab 6: Directory Management

1. Create Directory
2. List Directories
3. Delete Directory
4. Exit
Enter your choice:
1
Enter directory name to create:
os lab
Directory 'os lab' created successfully.

1. Create Directory
2. List Directories
3. Delete Directory
4. Exit
Enter your choice:
4
Exiting program...
geetha@Geetha:~/os$ nano lab7.sh
geetha@Geetha:~/os$ chmod +x lab7.sh
geetha@Geetha:~/os$ ./lab7.sh
Lab 7: Memory Usage Monitor

Current Memory Status (using free command):
               total        used        free      shared  buff/cache   available
Mem:           7.6Gi       509Mi       7.1Gi       3.5Mi       204Mi       7.1Gi
Swap:          2.0Gi          0B       2.0Gi

Detailed Memory Info (using top command):
MiB Mem :   7785.2 total,   7219.8 free,    510.6 used,    205.0 buff/cache

Available Memory:
Available: 7.1Gi
geetha@Geetha:~/os$ nano lab8.sh
geetha@Geetha:~/os$ chmod +x lab8.sh
geetha@Geetha:~/os$ ./lab8.sh
Lab 8: Environment Variables Demo

Local Variables:
```

```
Local Variables:
Name: Thenmozhi
Course: Operating Systems

Using printenv command:
Thenmozhi
Operating Systems

All Environment Variables (using env):
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
WT_SESSION=7304976e-d328-4004-a214-ec7a6f000b91
NAME=Geetha
PWD=/home/geetha/os
LOGNAME=geetha
HOME=/home/geetha
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/723_interop
WAYLAND_DISPLAY=wayland-0
TERM=xterm-256color
USER=geetha
DISPLAY=:0
SHLVL=2
XDG_RUNTIME_DIR=/run/user/1000/
WSLENV=WT_SESSION:WT_PROFILE_ID:
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Wi
ndows/System32/WindowsPowerShell/v1.0/:/mnt/c/Windows/System32/OpenSSH/:/mnt/c/Program Files/dotnet/:/mnt/c/Program Files/Git/cmd:/mnt/c/Users/Dell/AppData/Local/Programs/Python/Python314/
Scripts/:/mnt/c/Users/Dell/AppData/Local/Programs/Python/Python314/:/mnt/c/Users/Dell/AppData/Local/Programs/Python/Launcher/:/mnt/c/Users/Dell/AppData/Local/Microsoft/WindowsApps:/snap/bi
n
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
HOSTTYPE=x86_64
course=Operating Systems
PULSE_SERVER=unix:/mnt/wslg/PulseServer
WT_PROFILE_ID={3c2d458e-0403-5c08-aedd-909080ea585e}
OLDPWD=/home/geetha
name=Thenmozhi
_=/usr/bin/env
geetha@Geetha:~/os$ nano lab8.sh
geetha@Geetha:~/os$ chmod +x lab8.sh
geetha@Geetha:~/os$ ./lab8.sh
Lab 8: Environment Variables Demo
```

```
Using printenv command:
Geetha
Operating Systems

All Environment Variables (using env):
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
WT_SESSION=7304976e-d328-4004-a214-ec7a6f000b91
NAME=Geetha
PWD=/home/geetha/os
LOGNAME=geetha
HOME=/home/geetha
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/723_interop
WAYLAND_DISPLAY=wayland-0
TERM=xterm-256color
USER=geetha
DISPLAY=:0
SHLVL=2
XDG_RUNTIME_DIR=/run/user/1000/
WSLENV=WT_SESSION:WT_PROFILE_ID:
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Wi
ndows/System32/WindowsPowerShell/v1.0/:/mnt/c/Windows/System32/OpenSSH/:/mnt/c/Program Files/dotnet/:/mnt/c/Program Files/Git/cmd:/mnt/c/Users/Dell/AppData/Local/Programs/Python/Python314/
Scripts/:/mnt/c/Users/Dell/AppData/Local/Programs/Python/Python314/:/mnt/c/Users/Dell/AppData/Local/Programs/Python/Launcher/:/mnt/c/Users/Dell/AppData/Local/Microsoft/WindowsApps:/snap/bi
n
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
HOSTTYPE=x86_64
course=Operating Systems
PULSE_SERVER=unix:/mnt/wslg/PulseServer
WT_PROFILE_ID={3c2d458e-0403-5c08-aedd-909080ea585e}
OLDPWD=/home/geetha
name=Geetha
_=/usr/bin/env
geetha@Geetha:~/os$ nano lab9.sh
geetha@Geetha:~/os$ chmod +x lab9.sh
geetha@Geetha:~/os$ ./lab9.sh
Lab 9: I/O Redirection Demo
Content written to file1.txt
Content appended to file1.txt

Reading from file1.txt using input redirection:
Hello, this is OS Lab
Appending new line
```

```
            ├─systemd(348)───(sd-pam)(349)
            ├─systemd-journal(42)
            ├─systemd-logind(177)
            ├─systemd-resolve(107)
            ├─systemd-timesyn(108)───{systemd-timesyn}(157)
            ├─systemd-udevd(92)
            ├─unattended-upgr(209)───{unattended-upgr}(256)
            └─wsl-pro-service(181)─┬─{wsl-pro-service}(212)
                                   ├─{wsl-pro-service}(213)
                                   ├─{wsl-pro-service}(214)
                                   ├─{wsl-pro-service}(215)
                                   ├─{wsl-pro-service}(218)
                                   ├─{wsl-pro-service}(225)
                                   └─{wsl-pro-service}(228)
geetha@Geetha:~/os$ jobs
[1]-  Stopped                 sleep 30
[2]+  Stopped                 sleep 50
geetha@Geetha:~/os$ sleep 500
^Z
[3]+  Stopped                 sleep 500
geetha@Geetha:~/os$ sleep 300
^Z
[4]+  Stopped                 sleep 300
geetha@Geetha:~/os$ cat
^Z
[5]+  Stopped                 cat
geetha@Geetha:~/os$ jobs
[1]   Stopped                 sleep 30
[2]   Stopped                 sleep 50
[3]   Stopped                 sleep 500
[4]-  Stopped                 sleep 300
[5]+  Stopped                 cat
geetha@Geetha:~/os$ bg %2
[2] sleep 50 &
geetha@Geetha:~/os$ jobs
[1]   Stopped                 sleep 30
[2]   Done                    sleep 50
[3]   Stopped                 sleep 500
[4]-  Stopped                 sleep 300
[5]+  Stopped                 cat
geetha@Geetha:~/os$ fg %3
sleep 500
^Z
```

```
[5]+  Stopped                 cat
geetha@Geetha:~/os$ jobs
[1]   Stopped                 sleep 30
[2]   Stopped                 sleep 50
[3]   Stopped                 sleep 500
[4]-  Stopped                 sleep 300
[5]+  Stopped                 cat
geetha@Geetha:~/os$ bg %2
[2] sleep 50 &
geetha@Geetha:~/os$ jobs
[1]   Stopped                 sleep 30
[2]   Done                    sleep 50
[3]   Stopped                 sleep 500
[4]-  Stopped                 sleep 300
[5]+  Stopped                 cat
geetha@Geetha:~/os$ fg %3
sleep 500
^Z
[3]+  Stopped                 sleep 500
geetha@Geetha:~/os$ fg %5
cat
i am running in forground
i am running in forground
^Z
[5]+  Stopped                 cat
geetha@Geetha:~/os$ jobs
[1]   Stopped                 sleep 30
[3]-  Stopped                 sleep 500
[4]   Stopped                 sleep 300
[5]+  Stopped                 cat
geetha@Geetha:~/os$ kill %1

[1]   Stopped                 sleep 30
geetha@Geetha:~/os$ jobs
[1]   Terminated              sleep 30
[3]-  Stopped                 sleep 500
[4]   Stopped                 sleep 300
[5]+  Stopped                 cat
geetha@Geetha:~/os$
```